
Recognizing the predictive boundary of historically restricted physical knowledge

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Abstract

Scientific models have finite domains, established only once a successor theory explains the failure. Here we show that the predictive boundary of a body of physical knowledge can be tested from that knowledge alone. Only relations recorded before a chosen date, the knowledge horizon, are admitted, with no later formula and no example of a failure; we ask whether relations it cannot account for rank above those it can. At an 1899 horizon, a neural encoder trained only on pre-1900 laws and shown only the (x, y) points of each relation ranks 55 twentieth-century law families above nine held-out pre-1900 groups with an area under the ROC curve (AUROC) of 0.925; the same network before training gives 0.649. A second test, fixed in advance, extrapolates one of four simple forms fitted to part of each curve and reaches 0.888 against 11 held-out pre-1900 families without a trained network. On five historical measurements, the discrepancy with a named pre-1900 law, the incumbent, disappears under the successful successor prediction in four cases and under none of ten alternatives with the wrong shape or mechanism. The construction applies to any period with a dated corpus, a prediction to test, a measurement-error model and a calibration set.

1 Introduction

Physics advances by retiring descriptions that have stopped working. The Rayleigh–Jeans law, the Dulong–Petit law and the classical account of electrical resistance were each accurate across the range in which they were built and each failed outside it, and in each case the failure was explained only by a successor theory. The domain over which a model can be trusted is therefore something the field has established in retrospect.

Machine-learning systems for scientific data are built for other targets. They are optimized to interpolate within a sampled range, to predict forward, or to recover a compact governing expression from observations [1, 2]. Conformal prediction calibrates residuals [3], change-point methods locate distributional shifts [4], and Bayesian model discrepancy compares observations with a simulator [5]. Symbolic regression searches for a replacement expression once a discrepancy has already been found [6, 7].

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30 However, none of these operations takes a dated body of physical knowledge as the object
 31 under test. A residual is large relative to some fitted model, a change point is a shift in some
 32 distribution, and a recovered expression is a fit to some data; in each case the knowledge
 33 under test is implicit in the setup. Nor does an unusual curve shape identify the departure,
 34 because a departure from a named law may keep an ordinary functional form while a familiar
 35 mechanism may produce an unusual one.

36 We call the date that bounds a body of knowledge its knowledge horizon: only relations
 37 recorded before that date are admitted, and the pre-horizon law tested against a given
 38 dataset is the incumbent. Here we make the knowledge horizon an explicit input and test it
 39 directly with EPOCH (Epistemic Probe of Classical Horizons). Every test states in advance
 40 the laws it may fit, the error model, the data it fits, every data-dependent choice and the
 41 statistic and error rate it reports (Methods). It returns one of four verdicts: compatible
 42 over the measured range; localized incompatibility, with an estimate of where the incumbent
 43 stops fitting; whole-range incompatibility; or abstention.

44 Each test reports three kinds of evidence separately, never as a single averaged score.
 45 The learned component is a neural encoder trained only on pre-horizon relations, with no
 46 example of a departure, which scores a new curve by its distance from them. The four-form
 47 component finds the stretch of a curve, the interior, over which one of four simple forms
 48 (power, exponential, linear or constant) fits well, extrapolates that form, and measures how
 49 far the remaining points, the frontier, fall from the extrapolation. The theory-conditioned
 50 component compares the measurements with the incumbent prediction and calibrates the
 51 residuals against a null distribution simulated from that prediction on the same grid with
 52 the same errors.

53 We set the horizon at 1899, the eve of the replacement of the classical model set by quantum
 54 and relativistic physics and a date at which the classical corpus can be reconstructed from
 55 dated primary sources. Every relation admitted to training is traced to a dated primary
 56 source and assigned to a law family, the set of relations that share one law; training, validation
 57 and test data are divided by family, and no law appears on both sides. Training uses no
 58 later formula and no synthetic breakdown example.

59 We find that an encoder built this way ranks twentieth-century law families above pre-1900
 60 families withheld from training, and we find that the four-form component reaches the same
 61 ranking without any trained network. On five historical datasets we find that the discrepancy
 62 with the pre-1900 prediction disappears under the successor prediction in four of the five
 63 and under no wrong-shape or wrong-mechanism alternative. EPOCH therefore tests the
 64 predictive domain of a dated body of knowledge without being told where that domain ends.
 65 The same construction can be applied to any period for which a dated corpus, an admitted
 66 prediction, an observation model and a calibration set exist.

67 2 Results

68 2.1 The object of a boundary test

69 EPOCH returns one of four verdicts for a set of measurements and a stated set of laws it may
 70 fit: compatible over the measured range; localized incompatibility, with an estimate of where
 71 the incumbent stops fitting; whole-range incompatibility, with no identifiable boundary; or
 72 abstention, when the evidence is insufficient (Fig. 1). A fixed rule decides which component
 73 produces the verdict. The four-form component is used when the measurements contain a
 74 contiguous stretch, the interior, over which a pre-1900 law fits well. The theory-conditioned
 75 component is used otherwise, when an incumbent prediction and a model of the measurement
 76 errors are available. When neither condition is met the verdict is abstention.

77 We calibrate each component against its own reference distribution: the four-form and
 78 learned components against their scores on pre-1900 data, and the theory-conditioned
 79 component against a null distribution simulated from the incumbent prediction. The four-
 80 form component receives a point cloud, the (x, y) pairs of one relation with formula, units and
 81 labels removed, and nothing else. It fits the best of four generic forms (power, exponential,
 82 linear and constant) on the interior and extrapolates the fit to the frontier, the remainder

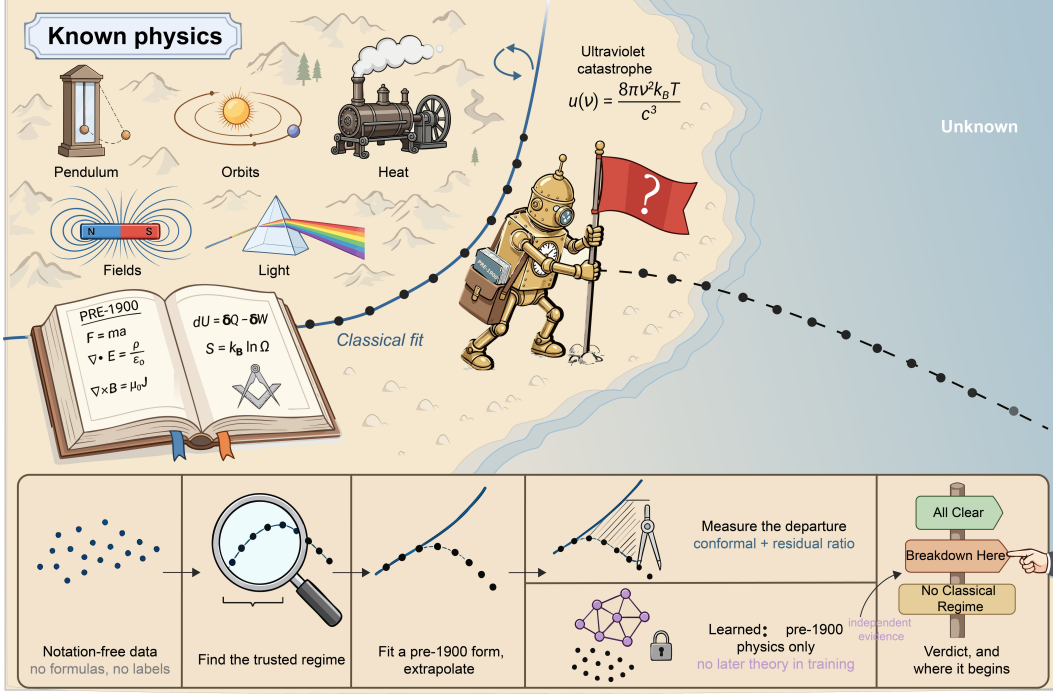


Figure 1. Testing a knowledge horizon. Top, an admitted classical form fitted inside the sampled regime (blue) tracks the measurements until they leave it; beyond that point the extrapolation and the data separate, and the question is whether the separation can be measured and located. Bottom, the test as applied here. A point cloud, a set of (x, y) pairs with no formula, unit or label, enters at the left. A fixed rule identifies the interior, the stretch over which an admitted pre-1900 form fits, and fits that form there; the extrapolation is compared with the frontier by two residual statistics (Methods); and an encoder trained on pre-1900 relations alone supplies an independent distance score. The signpost carries the verdict and, where an interior is present, the location at which compatibility is lost; a fourth output, abstention, applies when neither the four-form nor the theory-conditioned component can be run.

of the data. Two statistics measure the failure of the extrapolation. The split-conformal nonconformity is the mean absolute frontier residual divided by the 90th percentile of the interior absolute residuals, that is, a frontier residual scaled by the spread of the residuals where the fit is trusted. The extrapolation-residual ratio is the median absolute log-residual at the frontier divided by its interior value. Each denominator carries a small floor (Methods). In this component the best-fitting generic form stands in for the incumbent, and a localized boundary is where that form stops fitting; the theory-conditioned component tests the named prediction itself.

The theory-conditioned component computes the residuals of the measurements from the incumbent prediction, scaled by the stated measurement errors, and compares their size with the null distribution obtained by simulating data from that prediction on the same grid with the same errors. The learned component is a neural encoder trained only on point clouds from relations recorded before the horizon, with no example of a departure; its score is the distance of a new point cloud’s embedding from the embeddings of the training relations. Table 1 collects the principal quantitative results reported below.

2.2 Recognizing later physics from pre-1900 relations alone

To ask whether later physics can be ranked above pre-1900 physics from the shape of a relation alone, we trained a 20.9-million-parameter encoder (the large encoder) on 880,896 point clouds derived from 3,441 dimensionless relations $y = f(x)$, each a reduction of a pre-1900 law. During training, a second network that reads each formula’s expression tree produces a target embedding, and the point-cloud encoder, which is invariant to the order

Table 1. Principal quantitative results. Original controls are the nine held-out pre-1900 groups as originally generated; regenerated controls are the same groups regenerated with the sampling and noise process of the later families; the same-generator set is a set of later formulas generated by the same process as the pre-1900 corpus; metrological groups are calibration artefacts or instrument transfer curves. The 38 and 55 later families are each traced to a primary citation. Untrained denotes the same network with its initial random weights and no training steps. The large encoder is the 20.9-million-parameter model; the compact encoder is a small model trained on 205 source-verified pre-1900 point clouds; the generic-function control encoder is its twin trained on generic smooth expressions. A breakdown family is a set of synthetic curves built to depart from a classical form.

Analysis	Result
Large encoder, original controls, 38 families from 1901–1950 vs nine pre-1900 groups	AUROC 0.921 [0.833, 0.994]; untrained 0.589
Large encoder, original controls, 55 families from 1901–1996 vs nine pre-1900 groups	AUROC 0.925 [0.851, 0.982]; untrained 0.649
Large encoder, regenerated controls, 55 later vs nine pre-1900 groups	AUROC 0.857 [0.741, 0.945]; untrained 0.545
Large encoder, same-generator set of 1901–1950 formulas vs nine pre-1900 groups	AUROC 0.536 [0.424, 0.643]; untrained 0.594 [0.436, 0.756]
Four-form component, 55 later vs 11 pre-1900 families	Four-form AUROC 0.888 [0.800, 0.959]; compact encoder 0.669 [0.534, 0.798]
Compact encoder trained on pre-1900 laws vs generic-function control encoder, 12 breakdown and 12 classical families	Family AUROC 0.903 [0.729, 1.000] vs 0.375 [0.167, 0.618]
Four-form component, 200 pre-1900 formula datasets and 30 synthetic breakdowns, at the 0.5 screening threshold	AUROC 0.88
Four-form component at the calibrated threshold, 12 breakdown and 12 classical families	True-positive rate 0.271 [0.073, 0.500]; 0/96 false positives at the 5%-targeted threshold
Advancing the horizon on identical data grids	4/5 discrepancies with the pre-1900 prediction disappear under the successor prediction
Wrong-shape and wrong-mechanism controls	5/5 wrong-shape and 5/5 wrong-mechanism predictions remain rejected
Sixteen measured series, 5 post-1900 vs 9 classical or metrological groups	Learned AUROC 0.62 [0.29, 0.89]; four-form AUROC 0.78 [0.38, 1.00]; both intervals include 0.5
Boundary localization on 900 held-out simulations	90% interval coverage 91.3% [89.3%, 93.0%] (Wilson interval)
Simulated instrument effects at maximum severity	False-positive rate 0.40 for saturation; 0.63 for censoring; ≤ 0.11 for all other tested effects

of the points, is trained to match it. At inference the encoder receives a normalized point cloud and nothing else, with no formula, graph, units, date, source or anomaly label. Every scoring choice was fixed before any score on later physics was computed (Methods). We report the area under the ROC curve (AUROC), the probability that a randomly chosen later family scores above a randomly chosen pre-1900 group, with ties counted as one half, so that 0.5 is chance and 1 is complete separation; unless stated otherwise, brackets give 95% bootstrap intervals resampled over families or groups (Methods).

On the 38 families dated 1901–1950, each traced to a primary citation, the trained encoder ranks later physics above the nine held-out pre-1900 groups, each a set of law families sharing one mechanism, with AUROC 0.921 [0.833, 0.994] (Fig. 2a; Fig. 3a), whereas the same network with its initial random weights and no training steps (the untrained network) gives 0.589 (Fig. 3c). The 17 post-1950 families give 0.935 [0.823, 1.000] against 0.784 untrained, and the 55-family union 0.925 [0.851, 0.982] against 0.649. Each of the four distance components separates the 55-family union on its own, with AUROCs from 0.800 to 0.925 (Fig. 3d).

The later families were generated by a different sampling process from the pre-1900 training corpus, so sampling alone could in principle produce the separation. To test this we regenerated all nine pre-1900 control groups, the regenerated controls, with the point count,

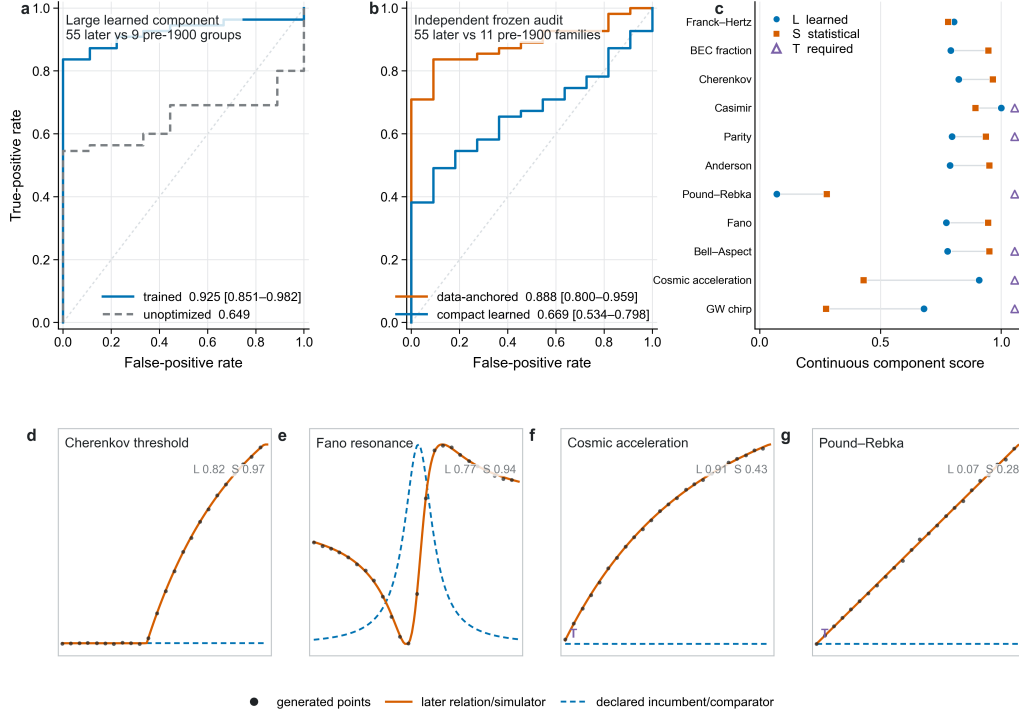


Figure 2. Temporal transfer and component scores across later physics. **a**, ROC curves for the large encoder trained on pre-1900 point clouds (blue) and the untrained network (grey), comparing 55 later law families with nine pre-1900 groups. **b**, Independent evaluation of the four-form component (orange) and compact encoder (blue) against 11 cited pre-1900 families. **c**, Learned scores (L, circles) and four-form percentiles against size-matched calibration (S, squares) for 11 further cases (Supplementary Note 17). Triangles mark cases assigned to the theory-conditioned component (T). **d–g**, Point clouds, later relations and incumbent predictions for Cherenkov emission [8], Fano resonance [9], cosmic acceleration [10] and Pound–Rebka redshift [11].

x-jitter, repeat count, noise range and heteroscedastic profile of the benchmark of 55 later law families described in the next subsection. AUROC is then 0.885 [0.778, 0.965] for the 38 families from 1901–1950 and 0.857 [0.741, 0.945] for all 55, against 0.516 and 0.545 for the untrained network (Fig. 3a,b), so within the 55-family benchmark the separation does not come from the difference in generating process alone. A second control, the same-generator set, generates the later formulas themselves, 1901–1950 formulas in this test, by the same process as the pre-1900 corpus (Supplementary Note 16). On that set the trained network gives 0.536 [0.424, 0.643] against 0.594 [0.436, 0.756] untrained: when the later formulas are drawn by the same generator as the training corpus, the trained encoder is near chance.

For the compact encoder, evaluated on a separate 192-dataset benchmark of 12 breakdown and 12 classical families, the gain is specific to the historical corpus. A breakdown family is a set of synthetic curves built by imposing one kind of departure, such as saturation, a step or a kink, on a classical form; such curves are used only for evaluation, never for training or for setting thresholds. A corpus reconstructed from cited pre-1900 sources supplies 268 point clouds from 51 cited pre-1900 law families, split by family into 205/17/46 training, validation and test datasets. A compact encoder trained on the 205 classical datasets (the compact encoder throughout, trained with no unit information) reaches dataset-level AUROC 0.854 by k-nearest-neighbour distance and 0.895 by shrinkage Mahalanobis distance, against 0.329 and 0.288 for matched random features, and family-level AUROC 0.903 [0.729, 1.000].

In contrast, a generic-function control encoder that we trained on generic smooth expressions, with identical architecture, seed, schedule, dataset count, family-size profile and functional-signature histogram (the elementary function types each formula contains), reaches 0.375 [0.167, 0.618]; the paired difference is 0.528 [0.264, 0.764]. Across three paired seeds the

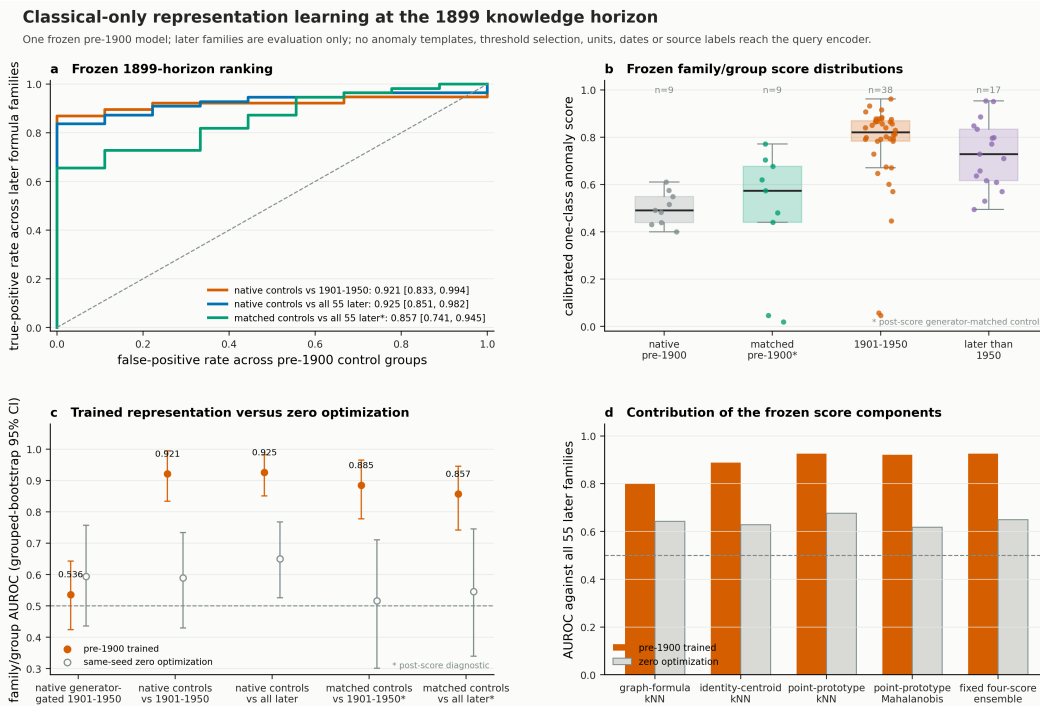


Figure 3. Large encoder at the 1899 horizon. **a**, The large encoder separates 38 families dated 1901–1950 and all 55 later families from nine pre-1900 groups. All-later AUROC is 0.857 [0.741, 0.945] against regenerated controls. **b**, Learned score distributions for original and regenerated pre-1900 controls and the two later periods. **c**, AUROCs for the trained and the untrained network. The same-generator comparison is shown separately. **d**, Each of the four distance components and their fixed equal-weight mean improves over the untrained network on the 55-family comparison. AUROCs and intervals are computed over law families or pre-1900 control groups.

two ranges are 0.889–0.910 and 0.292–0.403. At the threshold fixed on held-out families the encoder detects 2/12 breakdown families with no false positive among the classical controls, and this count is not stable across the three seeds (2/12, 8/12 and 1/12; Supplementary Note 8). A deformation control that retains each dataset’s physical units gives family-balanced AUROC 0.684 with unit information and 0.922 without it, so the ranking does not depend on unit information, and supplying it lowers the AUROC.

2.3 Explicit extrapolation as independent evidence

We next asked whether explicit extrapolation alone ranks later families above held-out pre-1900 families, without a learned representation. The 55-family benchmark contains 55 named physical-law families first valid from 1901 to 1996, each traced to a primary citation. Eight noisy 120-point samples of each give 440 formula-generated point clouds (Fig. 2b); eleven cited pre-1900 families excluded from encoder training are the controls. Each law family is scored by the median over its point clouds (Methods). The four-form component ranks later families above the earlier holdouts with AUROC 0.888 [0.800, 0.959], and the compact encoder with 0.669 [0.534, 0.798]. On a separate set of 200 pre-1900 formula datasets and 30 synthetic datasets from five breakdown families, with its decision rule set in advance, the four-form component gives AUROC 0.88. At the screening threshold of 0.5 (the median of the calibration scores, used only to select cases for further testing) recall is 0.90 and the false-positive rate is 0.24.

The ranking also varies with functional form. Learned and four-form AUROCs are 0.629/0.795 for 12 later affine, power or exponential laws, 0.600/0.982 for five threshold or piecewise families, and 0.909/0.864 for six nonmonotone or oscillatory families, so the two scores respond differently to smooth familiar forms, to thresholds and to nonmonotone curves.

When we apply both components to 11 further twentieth-century cases (Fig. 2c–g), the two components diverge again: Casimir plates [12] give 1.000/0.892, Cherenkov emission [8] 0.823/0.965, cosmic acceleration [10, 13] 0.908/0.429 and Pound–Rebka redshift [11] 0.071/0.277. Cases that report a single value or only a few points, including the Lamb shift [14] and the electron anomalous magnetic moment [15], are assigned to the theory-conditioned component or to abstention and are not part of this comparison (Supplementary Note 17). The calibrated detection rate of the four-form component at a fixed false-positive rate is reported below (Calibration, thresholds and the observation process).

2.4 Historical measurements at the 1899 horizon

We then scored four historical datasets that illustrate two of the four verdicts, whole-range incompatibility and localized incompatibility (Fig. 4a–d); a fifth, Millikan’s photoelectric measurement, is treated separately below. The four-form component’s initial screen gives percentiles (the fraction of size-matched classical calibration simulations scoring lower) of 0.95 for FIRAS, 0.95 for copper specific heat, 0.66 for Bertozzi and 0.66 for Onnes; whether an interior is present then determines which of the two outputs applies. Supplementary Note 4 gives finite-sample tail probabilities and the multiplicity calculation.

The COBE-FIRAS monopole spectrum of the cosmic microwave background [16] has 43 points and, at $T = 2.725$ K, samples $x = h\nu/kT$ from 1.2 to 11.3, entirely outside the regime $x \ll 1$ in which the Rayleigh–Jeans law [17, 18] holds. The incumbent prediction, intensity proportional to ν^2 , over-predicts the measured intensity above the peak by 475-fold to 3,000-fold, depending on how many of the lowest frequencies are used to fit the law. Fitting to the lowest 20% of frequencies gives a bootstrap interval of [1,500, 2,400]. No interior is present, and the verdict is whole-range incompatibility with the Rayleigh–Jeans prediction.

Bertozzi’s electron measurements [19] report kinetic energy against $\beta^2 = v^2/c^2$. The classical law $\frac{1}{2}mv^2$ crosses $\beta^2 = 1$ at 0.26 MeV and reaches $\beta^2 = 58.7$ at 15 MeV, equivalent to $7.7c$, whereas the measured β^2 values approach 1. The lowest point is already $0.87c$, placing the complete series outside a non-relativistic interior, and the output is whole-range incompatibility with the Newtonian energy–velocity prediction.

The Dulong–Petit law [20] predicts temperature-independent molar heat capacity, whereas the NIST cryogenic reference curves for six solids fall at low temperature. Five materials contain a high-temperature plateau, the interior, and support localized incompatibility relative to the constant model; beryllium gives whole-range incompatibility. Interior R^2 ranges from 0.965 to 0.998 and extrapolation-residual ratios from 127 to 1,490, and for oxygen-free high-conductivity copper the classical extrapolation over-predicts the 4 K reference value by a factor of 3.7×10^3 . The reference-polynomial curves share one low-temperature mechanism and are grouped accordingly.

Onnes’s mercury resistance values [21], digitized from $R(T)$ and anchored by the reported resistance below $10^{-5} \Omega$, show a smooth normal-state trend followed by a discontinuous drop. Relative to the smooth normal-state and residual-resistance models [22], EPOCH localizes a candidate boundary at 4.20 K, with figure-reading uncertainty and the censored lower values included in the observation model.

In Millikan’s 1916 photoelectric measurement, by contrast, the shape evidence and the theory evidence disagree [23]. Stopping potential is linear in frequency, $V_{\text{stop}} = (h/e)(\nu - \nu_0)$, with $R^2 = 0.996$ and a slope that recovers h/e to 0.3% (Fig. 4e), so the four-form component accepts the shape. On the same four measurements the theory-conditioned component rejects the classical prediction of frequency-independent photoelectron energy. Their fitted intercept gives $\nu_0 = 4.39 \times 10^{14}$ Hz, and because all four frequencies lie above ν_0 the output is whole-range incompatibility.

Conversely, independently measured Solar-System orbits recover Kepler’s exponent of $3/2$ [24] to four figures (1.500) and score 0.02, the four Galilean moons give exponent 1.500 and score 0.00, and Boyle’s original 1662 pressure–volume table [25] gives exponent -0.998 and scores 0.38 (Supplementary Fig. 1a–c). All three classical controls fall below the 0.5 screening threshold.

2.5 Advancing the horizon on the same measurements

We scored each dataset twice on the same grid and under the same error model: first under its incumbent prediction, then under the prediction of the theory that historically replaced it. We call this advancing the horizon. The pre-1900 prediction is rejected in all five cases (bootstrap $p = 0.001$; Methods; Fig. 5a). The successor prediction is not rejected in four of the five: FIRAS under Planck’s law [26] ($p = 1.000$), Bertozzi under relativistic kinetic energy [27] ($p = 0.089$), Onnes under a superconducting transition model ($p = 1.000$) and Millikan under the photoelectric relation [28] ($p = 0.900$), with discrepancy statistics falling by factors from 2.0×10^4 to 7.3×10^{10} .

Copper heat capacity, however, is the exception and is sensitive to the error model. A Debye-plus-electronic form [29] reduces the discrepancy by 3.86×10^3 but remains rejected under a 5% reference-curve error floor, the minimum relative error assumed for the reference polynomial ($p = 0.001$), and passes at 7.5%. Across all five cases, low-dimensional wrong-shape families remain rejected after Benjamini–Hochberg correction for the number of alternatives tested ($q \leq 0.0015$), as do dimensionally coherent wrong-mechanism forms (1,000 bootstrap replicates; $p = q = 1/1001$). These alternatives were chosen after the datasets were known, so this is a retrospective test (Supplementary Note 11). Four of the five discrepancies therefore collapse under the successful successor prediction and under none of the tested alternatives.

2.6 Measured series at the same horizon

To test both components on measured data, we scored sixteen NIST measurement series (Methods). The series fall into fourteen mechanism groups, each the set of series produced by one physical mechanism; five groups have a post-1900 mechanism and nine are classical or metrological, a metrological group being a calibration artefact or an instrument transfer curve. Groups are the units over which the AUROC is computed and resampled. For the five post-1900 groups against the nine others, the learned score gives AUROC 0.62 [0.29, 0.89] and the four-form score 0.78 [0.38, 1.00], and both intervals include 0.5 (Fig. 6 and Supplementary Note 18).

Copper thermal expansion measured from 24 K to 852 K has the highest post-1900 learned score; its coefficient falls towards zero below the Debye temperature, whereas classical equipartition of lattice modes gives a temperature-independent coefficient. The density profile of a cold Bose gas [30, 31] has the highest four-form score. A sulfur Rydberg series [32], whose inverse-square form was available empirically in 1888, belongs to the classical class yet ranks above every post-1900 group on the learned score (0.1588 against a highest post-1900 group value of 0.1586; Supplementary Note 18).

The circular-interference transmittance series rises to a sharp resonance at 451 nm, the most unusual shape in the collection. Its learned score is nonetheless low (0.0630), and its four-form statistic of 1.946 lies above the classical median of 0.566 but below the detection threshold of 3.391320708 (Supplementary Note 18). Superconducting magnetization relaxation (flux creep [33]), which no pre-1900 theory accounts for, is close to logarithmic in time and has the lowest four-form score of any series. Across the whole collection the two highest learned scores belong to a five-year standard-resistor record and a 23-point polymer-impedance curve.

2.7 Calibration, thresholds and the observation process

We compared the four-form component with two baselines on a 36-dataset synthetic benchmark on which every method receives the same inputs: it reaches AUROC 0.80 [0.64, 0.94], against 0.58 for Gaussian-process discrepancy [34] and 0.77 for split-conformal prediction (Supplementary Fig. 1d,f). On the benchmark phenomena of Supplementary Note 3 the component reaches AUROC 1.00 whether the classical law is supplied or selected automatically from the four forms, and 0.97 when the interior is also selected automatically.

We fixed the four-form threshold separately at $n = 5, 9, 43$ and 120 on classical calibration simulations and evaluated it over 12 breakdown and 12 classical families. At $n = 120$, the true-positive rate is 0.271 [0.073, 0.500] at the 5% threshold and 0 at the 1% quantile, with

an observed false-positive rate of 0/96 at both. Sensitivity is 1.0 for smooth roll-off and discontinuous-drop simulations and zero for seven families.

We then altered the observation process while holding the underlying classical relation fixed, to find where that specificity fails. At the strongest simulated levels a detector calibrated to 5% reports false-positive rates of 0% for calibration drift, 40% for sensor saturation, 63% for censoring, 9% for x-axis error, 11% for outliers, 2.5% for correlated noise and 0% for heteroscedasticity (Fig. 5b). Saturation and censoring are therefore tested explicitly for plateau and threshold cases.

We calibrate boundary intervals in the same way. Ordinary measurement bootstrap intervals cover only 53.9% of true boundaries at a nominal 90% level (358 simulations), and 8.5% for steps. Interval radii calibrated separately at each sample size on 899 independent simulations raise held-out coverage to 91.3% [89.3%, 93.0%] (Wilson interval) across 900 simulations, with family-wise coverage from 87.8% for smooth turnovers to 94.4% for kinks (Fig. 5c).

Finally, simulations extend the evaluation into interiors absent from the FIRAS and Bertozzi measurements. Blackbody radiation spanning $x = h\nu/kT$ from well below to well above unity contains a Rayleigh–Jeans interior with $R^2 = 0.99$ followed by a departure with split-conformal nonconformity 468 and extrapolation-residual ratio 192. Relativistic kinetic energy yields a localized boundary at $0.86c$, and simulated photoelectric voltage localizes the frequency threshold.

3 Discussion

We have introduced a test of whether a dated body of physical knowledge can account for a given set of measurements, and applied it at an 1899 horizon. Its distinguishing feature is that the horizon is an explicit input: training uses no formula recorded after 1899 and no example of a departure. Trained in this way, the large encoder ranks 55 twentieth-century law families above nine held-out pre-1900 groups with AUROC 0.925, against 0.649 for the untrained network, and 0.857 when the nine controls are regenerated with the 55-family benchmark’s sampling and noise process. The four-form component, which uses no trained network, reaches 0.888 against 11 held-out pre-1900 families on the same 55 later families. Ranking and detection are different quantities: on the synthetic benchmark of 12 breakdown and 12 classical families, at the calibrated 5% threshold, the four-form component detects breakdown datasets at a true-positive rate of 0.271 [0.073, 0.500] with 0/96 false positives.

EPOCH draws on neural extrapolation [1, 2], split-conformal prediction, Bayesian model discrepancy and change-point detection [3–5], and shares machinery with model-independent searches in particle physics[35] and with machine rediscovery of natural laws [6, 7, 36]. What differs is the target of inference: we test the predictive domain of a stated model before asking symbolic regression or theory construction for a replacement. Once a calibrated discrepancy exists, that search can proceed under a parsimony prior [37], as the Bayesian machine scientist and AI Feynman do in recovering Planck-type expressions from blackbody-type data [6, 7], and physical interpretation then supplies the mechanism behind the fitted form.

A discrepancy is interpretable only when it is tied to a named incumbent. When we score each historical dataset twice on the same grid and under the same error model, four of five old-horizon discrepancies disappear once the successor prediction is substituted, while wrong-shape and dimensionally coherent wrong-mechanism alternatives stay rejected after multiplicity correction. Copper heat capacity is the exception: the successor model passes only when the reference-curve error floor is relaxed from 5% to 7.5%, so the error floor has to be stated with the incumbent prediction. For a closed-form theory, advancing the horizon amounts to replacing the incumbent prediction and recalibrating its null distribution, and simulator-based references give the corresponding construction where no closed form exists. A conclusion about a named physical theory requires that the admissible laws and the observation model encode that theory’s prediction for the measured quantity (Methods).

Shape novelty and predictive failure are not the same quantity. Later affine, power and exponential laws often stay close to one of the four generic forms (learned and four-form AUROC 0.629/0.795). Threshold and piecewise relations separate on the four-form score

(0.982) but not on the learned score (0.600), and nonmonotone and oscillatory relations separate on both (0.909 and 0.864). Conversely, a discrepancy with a named incumbent can keep an ordinary shape, as Millikan’s stopping potential and the Pound–Rebka redshift both do. A classical circular-interference resonance scores 0.0630 on the learned component and stays below the four-form threshold, while superconducting flux creep [33], which no pre-1900 theory accounts for, is close to logarithmic in time and sits at the bottom of the four-form ranking. Keeping the component scores separate is what preserves this distinction between an unusual shape and the failure of a named prediction, and we expect it to matter most where the departure keeps an ordinary functional form.

The negative class of the 1899 experiment is small. Nine held-out pre-1900 groups carry the primary learned comparison and set the width of its interval, and fourteen mechanism groups set the width of the measured-series intervals. The learned ranking also depends on how the later formulas were generated. Regenerating the controls with the 55-family benchmark’s sampling process gives 0.857, but this analysis, added after the primary results were known, removes the sampling difference within that benchmark only. When the later formulas are instead generated by the same process as the training corpus, the trained encoder gives 0.536 [0.424, 0.643] (Results), close to chance. On measured series the ranking falls to 0.62 for the learned component and 0.78 for the four-form component, with intervals that span chance. The two highest learned scores in that collection belong to a standard-resistor record and a polymer-impedance curve, a pattern consistent with the observation process setting the largest learned-score differences on measured data. Whether this is the sensitivity that saturation and censoring expose for the four-form component in the simulated instrument tests remains to be tested.

Large libraries of measured relations could be screened against a fixed horizon, and selected cases rerun with explicit measurement errors and incumbent predictions to obtain a model test and, where an interior is sampled, a boundary estimate. Moving the horizon to another period requires a different dated corpus, admitted prediction, observation model and calibration set. The next experiment we plan is to fix the full decision procedure in advance and run it against measurements selected without regard to whether their physics is pre- or post-1900, and whose instrument models have been characterized.

4 Methods

4.1 Terms

Knowledge horizon: a calendar date; only physical relations recorded before it are admitted to the training corpus and to the laws a test may fit; here 1899. *Incumbent*: the pre-horizon law that a test examines for a given dataset; the incumbent prediction is its numerical prediction for the measured quantity. *Interior* and *frontier*: the contiguous stretch of a dataset over which one admitted pre-horizon form fits well, and the remainder, on which the extrapolation of that fit is tested. *Point cloud*: the (x, y) pairs of one relation, normalized to remove scale and stripped of formula, units, labels and ordering; the only input the learned and four-form components receive. *Law family* and *mechanism group*: the set of point clouds derived from one physical law traced to one dated source, and the set of measured series produced by one physical mechanism; both are the units over which AUROCs are computed and bootstrap intervals resampled; the nine pre-1900 control groups are sets of law families that share one mechanism.

Four-form component: fits the best of power, exponential, linear and constant forms on the interior, extrapolates to the frontier and reports the split-conformal nonconformity (the mean absolute frontier residual divided by the 90th percentile of the interior absolute residuals) and the extrapolation-residual ratio (the median absolute log-residual at the frontier divided by its interior value), each with a floor in its denominator; called the data-anchored or statistical component in the code and SI. *Theory-conditioned component*: computes the residuals of the measurements from the incumbent prediction, scaled by the stated measurement errors, and compares their size with a null distribution simulated from that prediction on the same grid with the same errors. *Learned component*: a neural encoder trained only on point clouds from pre-horizon relations, with no example of a failure; its score is the distance of a new

cloud’s embedding from those of the training relations; the large encoder has 20.9 million parameters and the compact encoder has width 128 and uses no unit information. *Null distribution*: the distribution of a discrepancy statistic over many datasets simulated from the incumbent prediction on the same x -grid and with the same error model as the real measurements. *Observation model*: the stated description of how the measurements were produced, comprising error distribution, censoring or saturation, and grid; an error floor is a minimum relative error assigned when no published covariance exists.

All thresholds, weights and scoring rules were fixed, and their files time-stamped and hashed, before any test data were scored; which component evaluates a dataset is decided by a rule fixed before scoring. *Verdicts*: compatible (the incumbent fits over the measured range); localized incompatibility (the incumbent fits an interior and fails beyond an estimated boundary); whole-range incompatibility (the incumbent fails everywhere sampled); abstention (the inputs do not support a test). *Advancing the horizon*: rescoring the same measurements on the same grid and with the same error model after replacing the incumbent prediction by the prediction of the theory that historically superseded it. *Regenerated and original controls*: the nine pre-1900 control groups regenerated with the sampling and noise process of the later families, and the same groups as originally generated; the same-generator set is a set of later formulas generated by the same process as the pre-1900 corpus. *Untrained network*: the same network with its initial random weights and no training steps.

4.2 Specifying a test

An EPOCH test specifies $\mathcal{T} = (\mathcal{M}, \Theta, P_\epsilon, I, A, S, \alpha)$. Here $\mathcal{M}$ lists the admissible models, Θ the fitted parameters, P_ϵ specifies the observation and censoring errors, I identifies the fitting data, A specifies every adaptive choice, S is the discrepancy statistic and α is the error rate. Because direction choice, window search, model selection and parameter fitting all belong to A , each of them is rerun inside every calibration and null replicate rather than being fixed once on the test data. Confirmatory tests use the finite-sample upper-tail probability $p = (1 + \#\{S_i \geq S_*\})/(n_{\text{cal}} + 1)$ with ties counted conservatively, and dataset scans report Benjamini–Hochberg adjusted q -values [38]. The main reported quantity is the true-positive rate at a threshold targeted to 5% false positives on independent calibration, with the 1% threshold secondary; the 0.5 threshold used in the historical screen, the median calibration score, selects cases for further testing. The finite-sample guarantee of the upper-tail test holds when calibration and test units are exchangeable; that unit is the law family (the source-law family of the SI) or the mechanism group, not the individual point or frontier.

4.3 The four forms and the interior and frontier

The four-form component draws on a fixed vocabulary of power, exponential, linear and constant forms. On validation data the four-form vocabulary extrapolates better than an eight-form one that adds quadratic, cubic, square-root and logarithmic terms. Given a point cloud, a fixed rule takes the candidate interior to be the contiguous regime in which one admitted form fits with high interior R^2 , and the frontier to be the complementary region. The search runs from both ends and fit quality decides whether the low- or the high-value direction is retained. The split-conformal nonconformity [39] is the mean absolute frontier residual divided by the 90th percentile of the interior absolute residuals, and the extrapolation-residual ratio compares the median log-residual at the frontier with its interior value. Each form is fitted from several initializations, and each denominator carries a floor, one percent of the median interior response magnitude for the nonconformity and 0.01 for the median interior log-residual, so that a near-perfect interior fit cannot produce an unbounded ratio. Relations with fewer than 20 points are calibrated against size-matched simulations.

4.4 Large-scale pre-1900 representation

The scaling experiment uses a 20,908,547-parameter encoder trained alongside a second, training-only network, without any anomaly supervision. Its query branch is a permutation-invariant six-block Set Transformer[40] of width 384 with eight heads, four pooling tokens and a 256-dimensional normalized output. During pretraining it is aligned to that expression-tree network, built from four edge-aware graph updates and two graph-local attention layers,

one of whose update networks is a 96-channel eight-basis radial-basis-function Kolmogorov–Arnold network module [41]. At query time the encoder receives an unordered (x, y) cloud normalized by median and interquartile range, and nothing else: no equation, graph, units, date, law name, post-cutoff label or deformation example.

Training draws on 880,896 clouds derived from 3,441 dimensionless relations $y = f(x)$, each a reduction of a pre-1900 law. These are grouped into 91 sets of clouds treated as the same law during training, spanning 64 law families. Held-out calibration and pre-1900 test sets contain 11,648 and 62,336 clouds from 91 and 487 formulas. Grouping the test-set formulas by mechanism gives the nine pre-1900 control groups used for every AUROC reported here (the historical-family groups of the SI). The model trains for 30 epochs at batch size 1,024, and the weights used for scoring are those selected by a rule fixed before any later-physics evaluation. Its learned score averages four calibrated ranks, each transformed by its own pre-1900 calibration empirical cumulative distribution function: top-eight cosine distance to graph-formula embeddings, top-eight distance to identity centroids, top-eight distance to point-formula prototypes, and shrinkage-Mahalanobis distance [42, 43] to those prototypes. Scores are summarized by the median over clouds within a formula and then within a law family or control group, and family- and group-level AUROCs use 2,000 grouped-bootstrap replicates; unless stated otherwise, bracketed intervals in this paper are 95% bootstrap intervals resampled at that level. The untrained network is scored by the identical procedure.

To build the regenerated controls we regenerated all nine pre-1900 control groups with the 55-family benchmark’s point count, x-jitter, repeat count, noise range and heteroscedastic profile, leaving the trained weights, the stored training embeddings, the calibration distribution function, score and aggregation unchanged. We then repeated the same analysis with the untrained network. A separate 96,064-cloud set of later formulas generated by the same process as the pre-1900 corpus, not taken from the 55-family benchmark, and a run with a 1950 horizon, are reported in Supplementary Note 16.

4.5 Compact encoder and its controls

The compact encoder is a set transformer of width 128 with three induced-set-attention blocks, and trains for 1,000 steps on two independently subsampled and rescaled copies of each of 205 pre-1900 point clouds from 40 families. Families are sampled without replacement so that repeated laws cannot become false contrastive negatives, and an auxiliary head predicts functional signatures. No breakdown, saturation, step, kink, splice or crossover example enters this optimization. We trained matched models with SI dimensions and with zeroed dimension metadata; the model trained without unit information is the one reported for the synthetic benchmark. The k-nearest-neighbour and shrinkage-Mahalanobis scores are compared against a matched random encoder. Leave-family-out calibration holds out each of the 40 training families in full, takes the median over the eight resampled copies of each dataset and then over the datasets within a family. The threshold, the 95th percentile of the 40 held-out family scores, is fixed before evaluation.

The generic-function control encoder is a second encoder trained without unit information that shares architecture, seed, batch size, augmentation, auxiliary objective and 1,000-step schedule but is trained on 205 point clouds from globally valid smooth expressions in 40 pseudo-families, matched to the historical set in dataset count, family-size profile and the histogram of eight-bit functional signatures, each bit recording whether one elementary function type (exponential, logarithm, sine, square root, power, inverse power, polynomial terms or hyperbolic) appears in the formula. We retrained both encoders from scratch under three paired initialization and augmentation seeds while the resampling seed and the 24 breakdown and classical families stay fixed, and we report the observed range and sample standard deviation across those runs. Finally, each of 46 held-out point clouds is paired with saturation, step, kink, splice, roll-off and crossover deformations that retain its physical-unit exponents. Each of seven law families contributes one undeformed and six deformed scores, the AUROC difference is bootstrapped by family, and each encoder is compared with its own threshold, the 95th percentile of its held-out family scores. Thirty-four of those datasets carry at least one nonzero dimensional exponent.

487 Which component scores a point cloud is fixed in advance. Point clouds with at least 20
 observations receive a learned score, point clouds whose response is nearly constant are sent 488
 instead to the theory-conditioned component, which tests a constant against noise, and 489
 formula and deformation families are the resampling units throughout. 490

4.6 Post-1900 formula benchmark 491

The 55-family benchmark contains 55 law families first valid between 1901 and 1996, each 492
 carrying a primary historical citation and 45 of them a DOI. The families include thermionic 493
 emission [44], discrete atomic excitation [45], parity violation [46], absence of diffusion in 494
 disordered lattices [47], the local-hidden-variable bound and its experimental test [48, 49] 495
 and the first gravitational-wave detection [50]. Every law is represented by a one-dimensional 496
 form $y = f(x)$ in dimensionless variables over a fixed support, sampled eight times at 120 497
 points with varying nuisance parameters, sampling and 0.5–2.5% noise, giving 440 datasets 498
 across 55 families. Special reductions, such as the BCS interpolation [51], the two-flavour 499
 neutrino survival curve [52] and the ideal quantum-Hall staircase [53], are noted in the 500
 benchmark listing. No benchmark family overlaps a family of the pre-1900 corpus. 501

The compact encoder, its calibration and its thresholds are those fixed in the leave-family-out 502
 evaluation above. The learned score is summarized by the median over resampled copies, 503
 datasets and families and compared with the threshold fixed on the 40 held-out family 504
 scores. The four-form score reruns the full four-form selection and extrapolation procedure 505
 against the existing $n = 120$ thresholds. Eleven cited pre-1900 families are the controls, 506
 AUROC intervals resample families, and twelve of the post-1900 laws deliberately overlap 507
 the affine, power or exponential forms. Component union and intersection rates are reported 508
 in Supplementary Note 15, and per-family conformal p -values are in the data package. The 509
 11 further twentieth-century cases were assembled after the primary scores were known 510
 (Supplementary Note 17). 511

4.7 Historical measurements and horizon-specificity controls 512

We analyse the published COBE-FIRAS monopole spectrum with its pointwise errors, five 513
 Bertozzi values digitized from the 1964 report, NIST specific-heat reference polynomials 514
 for six materials, Onnes resistance values including upper limits, Millikan sodium stopping 515
 potentials and Solar-System orbital constants. The incumbent predictions are the classical 516
 predictions that follow from physics available before 1900; where the standard statement of 517
 a prediction was published later, as for the Rayleigh–Jeans form in 1900 and 1905 and the 518
 Drude model in 1900, we cite that statement as the reference and do not use it as a training 519
 source. For each case we record the model set, fitted parameters, observation model, fit 520
 subset, adaptive choices, decision rule and output. We evaluate old- and advanced-horizon 521
 null distributions on the same x -grid; explicit error floors stand in for unavailable covariance 522
 and for digitization uncertainty, and we vary them in sensitivity analysis. 523

To establish that a collapse in the discrepancy is specific to the successful successor, five 524
 wrong-shape forms, one per case, and five dimensionally coherent wrong-mechanism forms 525
 are refitted on the same case grids and observation models with 1,000 parametric-bootstrap 526
 replicates. The physical controls substitute fermionic occupation for photon statistics, a 527
 proton for the electron rest-energy scale, an Einstein oscillator [54] for Debye acoustic 528
 modes [29], Arrhenius transport for the superconducting transition, and Compton recoil for 529
 the photoelectric slope. 530

4.8 Measured series 531

We took sixteen series unmodified from the NIST Dataplot and Statistical Reference Dataset 532
 collections. For each series we specified, before any score was computed, the data columns, 533
 the 1899 incumbent, the mechanism group and whether the mechanism was known before 534
 1899. Series are sorted by abscissa and thinned uniformly to at most 2,000 points. The 535
 learned component applies the fixed rule that sends nearly constant responses to the theory- 536
 conditioned component, rescales each measured series and adds no synthetic noise, with 537
 the stored training embeddings computed from the pre-1900 training set. The four-form 538

539 component reruns the complete four-form direction, window, model-selection and scoring
540 procedure on eight resampled copies of each series at the largest calibration sample size the
541 series supports. Scores are summarized by the median within a series and then within a
542 mechanism group, and the grouped AUROC resamples mechanism groups.

543 4.9 Synthetic suites, baselines and localization

544 The held-out evaluation comprises 180 labelled datasets and the fair baseline suite 36, on which
545 split-conformal prediction, the extrapolation-residual ratio, Gaussian-process discrepancy [34],
546 CUSUM [4] and RESET [55] are compared. The mechanism-level benchmark contains 12
547 classical and 12 breakdown families with eight datasets each; its 5% and 1% thresholds
548 come from 785 valid size-matched calibration simulations, and uncertainty is resampled by
549 breakdown or classical family. The learned-component evaluation uses 192 datasets over the
550 same 24 families. Instrument tests apply calibration drift, sensor saturation, censoring, x-axis
551 error, outliers, correlated noise and heteroscedasticity to classical synthetic relations while
552 leaving the underlying law intact, and boundary intervals are obtained by split-conformal
553 calibration of the normalized localization error separately at $n = 40, 80$ and 160.

554 4.10 Measured-series collection

555 We downloaded forty-four files from the NIST Dataplot and Statistical Reference Dataset
556 collections. The collection holds 20 instrument controls, 11 classical controls, six controls
557 whose post-1900 mechanism is known and seven horizon cases. The sixteen series analysed
558 are listed in Supplementary Note 18.

559 5 Data availability

560 The public COBE-FIRAS, Solar-System, Bertozzi, NIST, Millikan and Kamerlingh Onnes
561 inputs are identified in the Supplementary Information. The data package contains
562 source URLs and hashes; the 55-family benchmark and its 440 generated point clouds;
563 the 11 pre-1900 control families and their 88 generated datasets; all 128 individual
564 scores of the measured series; the pre-1900 training, calibration and evaluation mani-
565 fests; and every numerical result used in the figures. The project repository is avail-
566 able at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>,
567 and the archive for this release at <https://doi.org/10.5281/zenodo.22742368>; <https://doi.org/10.5281/zenodo.22138620> resolves to the most recent version.

569 6 Code availability

570 Code is available at <https://github.com/ChenxiHeCam/Epistemic-probe-of-Classical-Horizons>.
571 The repository contains corpus checks, point-cloud generators, evaluators, statistical calibration and robustness experiments,
572 model definitions, training configurations, figure scripts and the validation scripts.
573 Checkpoint and data hashes are verified before evaluation, and the release archived
574 at <https://doi.org/10.5281/zenodo.22742368> is the version used for every number
575 reported here.

577 7 Acknowledgements

578 We thank NASA LAMBDA, the NIST Cryogenic Material Properties database and the
579 KNAW digital archive for public data access. This research received no specific grant from
580 any funding agency in the public, commercial, or not-for-profit sectors.

581 **8 Author contributions**

C.H. conceived the framework, designed and built EPOCH, ran the experiments, and wrote 582
the manuscript. J.C. contributed to the benchmark construction, case-study analysis, and 583
manuscript revision. Both authors reviewed and approved the final manuscript. 584

9 Competing interests 585

The authors declare no competing interests. 586

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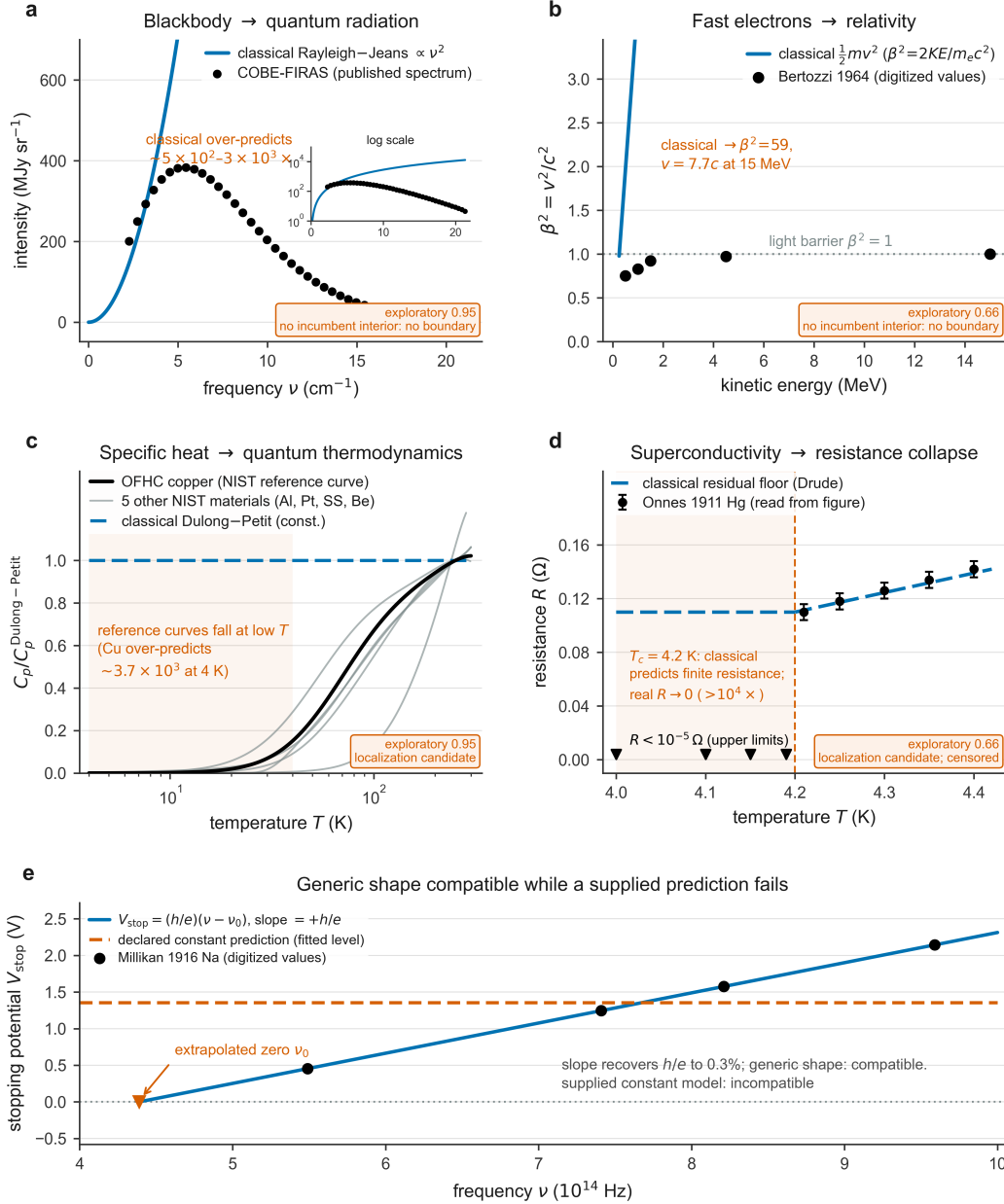


Figure 4. Historical measurements under the 1899 horizon. a–d, Black points show published processed measurements, digitized historical values or evaluated reference curves as labelled; blue curves show incumbent predictions. Orange badges give screening percentiles. FIRAS and Bertozzi yield whole-range incompatibility. Specific heat and Onnes contain an interior, allowing a candidate boundary to be marked by the shaded frontier. Tail probabilities and multiplicity calculations are reported in Supplementary Note 4. e, In Millikan’s 1916 digitized values, stopping potential rises linearly with frequency as $V_{stop} = (h/e)(\nu - \nu_0)$. The slope recovers h/e to 0.3%, and the marker denotes the extrapolated zero crossing ν_0 . The four-form score falls below the threshold, whereas the theory-conditioned component rejects the classical prediction of frequency-independent energy (dashed line).

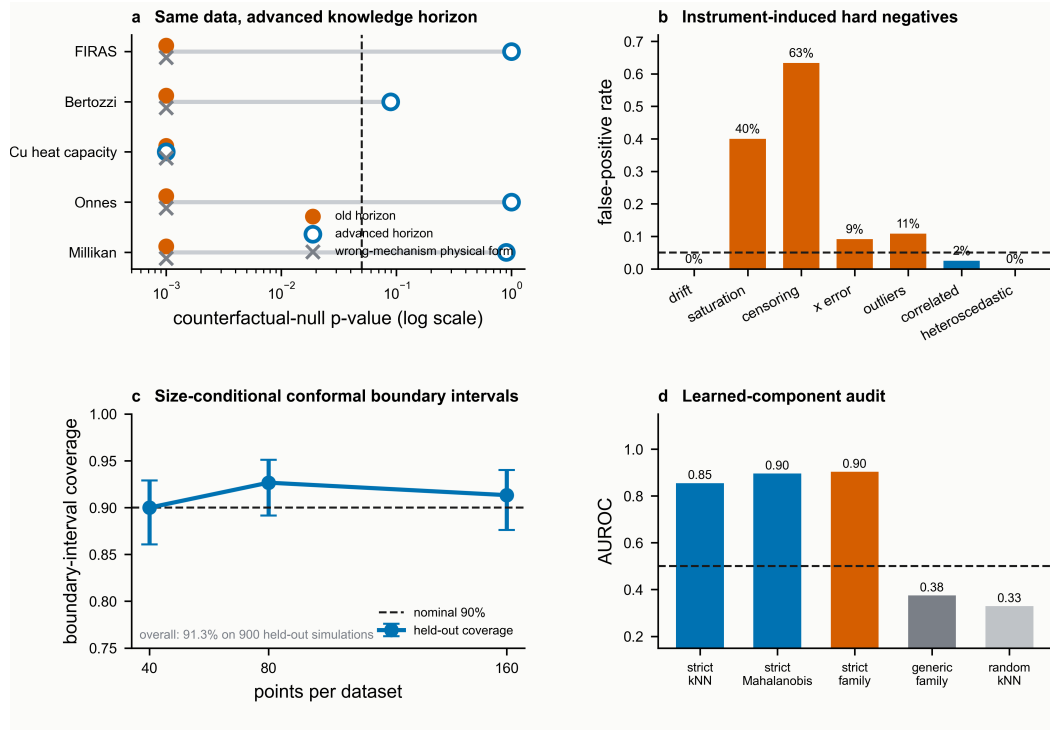


Figure 5. Calibration and controls for the EPOCH components. **a**, Goodness-of-fit p -values on the same measurement grid under the pre-1900 prediction and its successor. Four of five successor predictions pass at 5%; five wrong-mechanism forms remain rejected. **b**, False-positive rates under simulated instrument effects, with the largest changes caused by saturation and censoring. **c**, Held-out coverage of 90% boundary intervals calibrated separately at each sample size. **d**, Compact encoder trained only on 205 datasets from 40 pre-1900 families with no breakdown example. Family-grouped AUROC is 0.903 [0.729, 1.000] against 0.375 [0.167, 0.618] for the generic-function control encoder on the same evaluation families; dataset-level AUROC is 0.854 by k-nearest-neighbour distance and 0.895 by shrinkage Mahalanobis distance, against 0.329 and 0.288 for matched random features (Results).

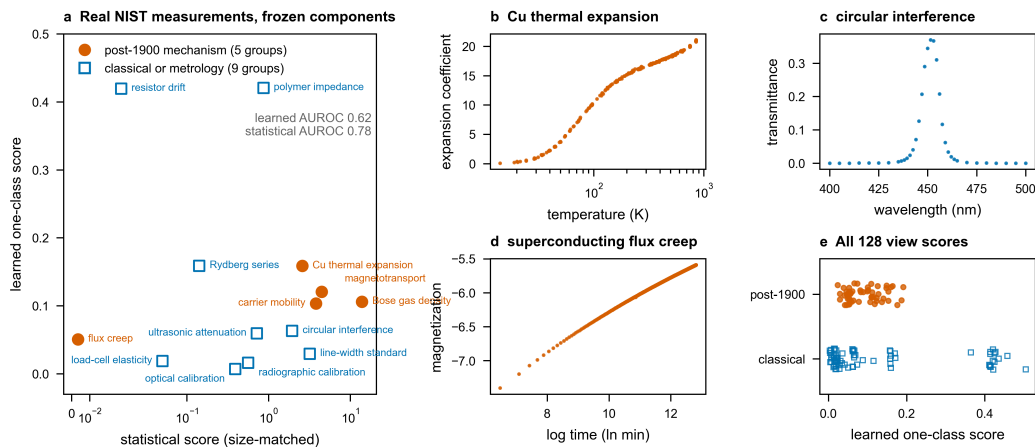


Figure 6. Measured series at the 1899 horizon. **a**, Mechanism-group medians of the learned and four-form scores for sixteen NIST measurement series in fourteen groups, five outside the pre-1900 model set and nine classical or metrological. **b–d**, Three of the measured series: copper thermal expansion from 24 K to 852 K, circular-interference transmittance, and superconducting magnetization relaxation against log time. **e**, Every individual score (eight resampled copies per series) behind the medians in **a**.

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